

Aylar Babaei

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PROFESSIONAL SUMMARY

Bioinformatics Scientist with 2+ years running and troubleshooting genomic pipelines (RNA-seq, DNA sequencing, multi-cohort harmonization) for translational genomics research. Builds HLA typing and neoantigen prediction workflows and provisions HPC/cloud infrastructure alongside senior bioinformatics staff. Brings a systems-thinking, engineering-minded approach to genomics work, with a growing focus on pipeline architecture and infrastructure design.

CORE SKILLS & TOOLS

Bioinformatics: RNA-seq, DNA sequencing (WGS/WES), multi-cohort data harmonization, HLA typing, neoantigen prediction

Bioinformatics Tools & Data: Familiarity with GATK, gnomAD, ClinVar, OMIM, UCSC Genome Browser, NCBI; FASTQ, BAM, CRAM, VCF, MAF, IDAT data formats

Scientific & Analytical: UNIX/Linux, Python, R, Bash/Shell, SQL; biological interpretation of genomic/transcriptomic data; results validation and benchmarking; cross-disciplinary collaboration with clinicians and bench scientists

Cloud & Workflow Engineering: CAVATICA/Seven Bridges platform, CWL, Docker, AWS (S3, EC2), HPC (VM provisioning, job scheduling), performance profiling and optimization, Git/GitHub (version control, PR collaboration)

Data Operations: Metadata harmonization, manifest generation, schema validation, QC automation, SOP documentation, data warehousing, basic SQL and relational table structures

EXPERIENCE

Bioinformatics Scientist

Jun 2024 – Present

Children's National Hospital, BTI Bioinformatics Core • Washington, DC

- Run multi-cohort genomic data harmonization workflows on raw RNA-seq/DNA sequencing data alongside two senior bioinformatics staff; debug and resolve pipeline errors, rerun failed jobs, and document fixes to support reproducibility.
- Provision and manage HPC/cloud compute environments (VMs, job scheduling); profile and optimize pipeline performance (runtime, memory, resource allocation).
- Built HLA typing and neoantigen prediction workflows (RNA2HLA, arcashHLA, IEDB-based MHC I binding) for immunogenomics analyses.
- Use AWS cloud compute (S3, EC2) and the CAVATICA/Seven Bridges platform to support execution of genomic data pipelines.
- Contribute to production pipeline pull requests, including methylation-based rare tumor subtyping within a national genomics consortium, under senior bioinformatics staff direction, using Git/GitHub for version control.

Graduate Research Assistant

Jan 2023 – May 2024

GWU Computational Biology Institute, Rahnavard Lab • Washington, DC

- Profiled and quantified LC/MS metabolomics data from iHMP studies to support massSight software development, including quality assessment and biological interpretation of high-dimensional datasets.

Graduate Lab Assistant

Mar 2023 – Dec 2023

GWU Cancer Center • Washington, DC

- Supported translational cancer biology studies on CDKN2A/P16 deletion and KRAS activation in mouse intestinal epithelial cells; assisted with Visium spatial transcriptomics generation, tracking, IHC workflows, and genotyping.
- Managed a database of over 2,000 laboratory mice, ensuring data accuracy and integrity; maintained breeding, genotype, and health records and assisted with breeding, weaning, and IP injections in support of the mouse model study.

EDUCATION

M.S. Bioinformatics, The George Washington University

Aug 2022 – May 2024

B.S. Chemistry, Shiraz University

Sep 2014 – May 2018